

Zhu Enze

Undergraduate Student

Intelligent Systems, Software Engineering, and Embedded Development

Xi'an Jiaotong University | Xi'an, China | jsxzznz@163.com | +86 13092344080

Homepage | GitHub | Google Scholar | CV PDF

Profile

I am Zhu Enze, a 2025 undergraduate in Automation at Xi'an Jiaotong University and a member of the 2023 Young Gifted Program, interested in intelligent systems, machine learning, software engineering, and embedded development.

Intelligent systems • Machine learning • Software engineering • Embedded development

Education

Xi'an Jiaotong University Sept. 2025 - Present

Bachelor of Engineering in Automation

Full undergraduate training in automation, control, software, and intelligent systems. | Programming in C (100/100) | Principles of Automatic Control | Engineering Mathematics

Xi'an Jiaotong University Sept. 2023 - Jun. 2025

Young Gifted Program (Preparatory Phase)

Accelerated preparatory training with early exposure to research, project practice, and competition-based learning. | Early research training | Project-based learning

Selected Projects

CoS-VRL: Step-Level Verifiable RL for Video Diffusion Reasoning Oct. 2026

Research Collaboration | Video diffusion models, Reinforcement learning, Verifiable rewards, PyTorch

- Estimate each denoising step's contribution to the final reasoning outcome with verifiable rewards.
- Reinforce only the genuinely critical denoising stages, improving reasoning accuracy while reducing training cost.

AI-Assisted Laparoscopic Surgery System Oct. 2026

Research Project | Computer vision, Surgical video analysis, Deep learning

- Contributing to the research groundwork for a laparoscopic-video-based surgical assistance system.

Chuang: Open-Source Linux Sky Window Sep. 2026

Open Source Project | Python, GTK4, libadwaita, Cairo, Open-Meteo

- Built offline astronomy and shadow calculations, real-time weather integration, wallpaper and always-on-top modes, and a four-language UI.
- Released under the MIT license with CI-built .deb packages.

RACDataset Fitness Video Understanding Apr. 2026

Research Project | Python, Computer vision, Video processing

- Built the data pipeline and designed a video segmentation algorithm for a 20,000+ video dataset.
- Dataset pipeline and segmentation algorithm delivered; follow-up research ongoing.

AI-Based Forest Fire Prediction Model 2025

AIC Competition Project | Python, PaddlePaddle

- Prepared structured meteorological inputs for CNN and LLM training and contributed innovations based on the YingLong model.
- Won a provincial third prize in the AIC competition.

Awards

Third-Class Scholarship · Top 13%, XJTU Scholarship (2025-26) Oct. 2026

Ranked 23/185 (top 13%) in the major for the 2025-2026 academic year.

University-Level Silver Award, China International Innovation Competition Aug. 2026

Team member on label-corrected cervical cytology image diagnosis.

Honorable Mention, Mathematical Contest in Modeling (MCM) 2026 May 2026

Team member alongside Li Haoxuan and Shi Chengqi.

Skills

Programming Languages: Python (data processing and machine learning), C / C++ (fundamental development), JS&TS / HTML / CSS / Astro (front-end development), Bash (Linux operations)

Frameworks & Tools: TensorFlow / PyTorch / PaddlePaddle, GTK4 and libadwaita (Linux desktop), Linux, SSH, Git, Firebase, LaTeX, Jekyll and static-site workflows

Hardware & Platforms: Arduino development fundamentals, Orange Pi deployment and maintenance, Practical Linux server operation

Languages: Chinese (Native), English (CET-6, proficient in reading academic literature)